

ICTM_Module_model_summary.txt

TBD2 pipeline output - rendered to PDF

NOTE ON THE SECTION BELOW: 'Overall Risk', the 'Risk Scenarios' ranking, and 'Follow-up Actions' are computed from randomly simulated stress-test scenarios (seeded for reproducibility), not from statistical evidence measured in this client's actual data. Treat them as illustrative stress-test projections, not as data-driven findings. (Model Evaluation Metrics further below ARE fit on the real data and are not affected by this note.)

Overall Risk: 0.38030394934865536

Risk Scenarios in Higher-First Priority:

Scenario_10: 0.9582314936921804

Scenario_9: 0.8038222834527426

Scenario_8: 0.6386992184153006

Scenario_7: 0.4897011323447945

Scenario_6: 0.3610822487815188

Scenario_5: 0.2517311237794721

Scenario_4: 0.15915817873087432

Scenario_3: 0.0901217409995456

Scenario_2: 0.03943721790523324

Scenario_1: 0.011054855384891085

Average MSE across models is 8.397562861146866e-06, indicating strong confidence in the risk assessment.

Follow-up Actions:

- High risk in Scenario_10 (Avg Risk=0.958) driven by metrics: LCP10102, RTS10101, CIP10107. Suggested mitigation: prioritize monitoring and corrective actions for these metrics.
- High risk in Scenario_9 (Avg Risk=0.804) driven by metrics: CIP10110, CIP10105, IOP10108. Suggested mitigation: prioritize monitoring and corrective actions for these metrics.
- High risk in Scenario_8 (Avg Risk=0.639) driven by metrics: LCP10111, RTS10102, DDD10106. Suggested mitigation: prioritize monitoring and corrective actions for these metrics.

Metrics involved in High-Risk Scenarios:

Scenario_10 (Avg Risk=0.958):

- LCP10102: Effort Variance (% Actual vs. Estimated)
- RTS10101: CAR (Causal Analysis and Resolution)
- CIP10107: Aging of Open Quality Issues

Scenario_9 (Avg Risk=0.804):

- CIP10110: Information Security
- CIP10105: Process Quality Assurance
- IOP10108: Innovation Effectiveness and Productivity

Scenario_8 (Avg Risk=0.639):

- LCP10111: Mitigation Plans Active vs. Completed
- RTS10102: RCA Pareto
- DDD10106: Design Review Defect Density

Baseline Predictor (Mean of PRCY_001):

MSE: 0.5295764126222222

R-squared: 0.0

RMSE: 0.7277200097717681

MAPE: 0.6725930594150659%

PyTorch Model Evaluation:

MSE: 9.163978088355295e-05

R-squared: 0.9998269564529323

RMSE: 0.009572866910364573

MAPE: 0.008714709588954662%

Model Evaluation Metrics:

MSE: 9.163978088355295e-05

RMSE: 0.009572866910364573

R²: 0.9998269564529323

MAPE: 0.008714709588954662%

Risk Confidence Score: High

Recommendation - Continuously monitor performance metrics and adjust strategies as needed.

Recommendation - Reassess risk management plans periodically to ensure effectiveness.

-----End of ICTM Module Risk Analysis Summary-----